

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Date": [
        "2025-08-01", "2025-08-02", "2025-08-03",
        "2025-08-04", "2025-08-05", "2025-08-06",
        "2025-08-07", "2025-08-08", "2025-08-09",
        "2025-08-10"
    ],
    "Stock Price": [100, 102, 101, 105, 108, 107, 110, 115, 113, 118]
}

# Convert into DataFrame
df = pd.DataFrame(data)

# Convert Date column into datetime
df["Date"] = pd.to_datetime(df["Date"])

df["Moving Average"] = df["Stock Price"].rolling(window=3).mean()

highest_price = df["Stock Price"].max()
lowest_price = df["Stock Price"].min()

peak_day = df[df["Stock Price"] == highest_price]
drop_day = df[df["Stock Price"] == lowest_price]
```

```
print("Highest Stock Price:\n")
```

```
print(peak_day)
```

```
print("\nLowest Stock Price:\n")
```

```
print(drop_day)
```

```
volatility = df["Stock Price"].std()
```

```
print("\nStock Volatility:", volatility)
```

```
plt.figure(figsize=(10,5))
```

```
# Stock Price Line
```

```
plt.plot(df["Date"], df["Stock Price"],  
         marker='o', label="Stock Price")
```

```
# Moving Average Line
```

```
plt.plot(df["Date"], df["Moving Average"],  
         linestyle='--', label="Moving Average")
```

```
# Labels and Title
```

```
plt.title("Stock Price Time-Series Analysis")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Price")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

